WORKED SOLUTION

STEP I 2002, Question 7

Pure • Trigonometric Integrals

STEP I 2002, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q7

The key idea

Neither I nor J looks friendly alone, but their sum and difference both are. Adding them puts the denominator on top, so the integrand is 1. Subtracting them puts the *derivative* of the denominator on top, so the integral is a logarithm. The two later integrals use the same trick in a more general form: split the numerator into a multiple of the denominator plus a multiple of the denominator's derivative.

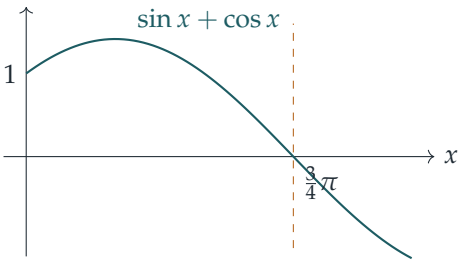
Solution

The main result

First, why the restriction on a? We can write

$$\sin x + \cos x = \sqrt{2}\,\sin\left(x + \tfrac{\pi}{4}\right),$$

which is positive for $-\frac{\pi}{4} < x < \frac{3\pi}{4}$ and first vanishes at $x = \frac{3}{4}\pi$. So for $0 \le a < \frac{3}{4}\pi$ the denominator is positive on all of $[0, a]$. Both integrals are then perfectly ordinary, and the logarithm below is of a positive number.



Adding the two integrals,

$$I + J = \int_0^a \frac{\cos x + \sin x}{\sin x + \cos x}\,\mathrm{d}x = \int_0^a 1\,\mathrm{d}x = a.$$

Subtracting them,

$$I - J = \int_0^a \frac{\cos x - \sin x}{\sin x + \cos x}\,\mathrm{d}x.$$

The numerator $\cos x - \sin x$ is exactly the derivative of the denominator $\sin x + \cos x$, so the integrand has the form $f'(x)/f(x)$ with $f > 0$ on $[0, a]$. Hence

$$I - J = \Big[\ln(\sin x + \cos x)\Big]_0^a = \ln(\sin a + \cos a) - \ln 1 = \ln(\sin a + \cos a).$$

Adding the two results gives

$$2I = a + \ln(\sin a + \cos a),$$

as required. (Subtracting them gives $2J = a - \ln(\sin a + \cos a)$ as a bonus.)

Part (i)

Since $p, q > 0$ and both $\sin x$ and $\cos x$ are non-negative on $[0, \frac{\pi}{2}]$ and never zero together, the denominator $D(x) = p\sin x + q\cos x$ is positive there. Its derivative is $D'(x) = p\cos x - q\sin x$.

Ask yourself How do we generalise "consider $I + J$ and $I - J$" when the denominator has coefficients p and q?

Before, the sum and difference worked because the numerators added up to the denominator and subtracted to its derivative. So now we look for constants A and B with

$$\cos x = A\,(p\sin x + q\cos x) + B\,(p\cos x - q\sin x) \qquad \text{for all } x.$$

Comparing coefficients of $\sin x$ and $\cos x$:

$$pA - qB = 0, \qquad qA + pB = 1.$$

The first gives $B = pA/q$; then the second gives $qA + p^2A/q = 1$, so

$$A = \frac{q}{p^2 + q^2}, \qquad B = \frac{p}{p^2 + q^2}.$$

Now the integral splits into an easy piece and a logarithmic piece:

$$\begin{aligned}\int_0^{\pi/2} \frac{\cos x}{p\sin x + q\cos x}\,\mathrm{d}x &= \int_0^{\pi/2} \left(A + B\,\frac{D'(x)}{D(x)}\right)\mathrm{d}x \\ &= \frac{\pi}{2}A + B\Big[\ln D(x)\Big]_0^{\pi/2} = \frac{\pi}{2}A + B\ln\frac{p}{q},\end{aligned}$$

using $D(\frac{\pi}{2}) = p$ and $D(0) = q$, both positive. Therefore

$$\int_0^{\pi/2} \frac{\cos x}{p\sin x + q\cos x}\,\mathrm{d}x = \frac{\frac{1}{2}\pi q + p\ln(p/q)}{p^2 + q^2}.$$

Part (ii)

Now the denominator is $D(x) = 3\sin x + 4\cos x + 25$, with $D'(x) = 3\cos x - 4\sin x$. The denominator is positive, since $|3\sin x + 4\cos x| \leq 5 < 25$. The constant 25 in D means we must also match the constant term, so we seek A and B with

$$\cos x + 4 = A\,(3\sin x + 4\cos x + 25) + B\,(3\cos x - 4\sin x).$$

Comparing coefficients:

$$\sin x:\ 3A - 4B = 0, \qquad \cos x:\ 4A + 3B = 1, \qquad \text{constant}:\ 25A = 4.$$

The first two give $A = \frac{4}{25}$ and $B = \frac{3}{25}$, and then the constant equation $25 \cdot \frac{4}{25} = 4$ is satisfied too. With three equations and only two unknowns, this is not automatic: the "+4" in the numerator was chosen to make it work. So

$$\begin{aligned}\int_0^{\pi/2} \frac{\cos x + 4}{3\sin x + 4\cos x + 25}\,\mathrm{d}x &= \int_0^{\pi/2} \left(\frac{4}{25} + \frac{3}{25}\,\frac{D'(x)}{D(x)}\right)\mathrm{d}x \\ &= \frac{4}{25}\cdot\frac{\pi}{2} + \frac{3}{25}\Big[\ln D(x)\Big]_0^{\pi/2} \\ &= \frac{2\pi}{25} + \frac{3}{25}\ln\frac{28}{29},\end{aligned}$$

since $D(\frac{\pi}{2}) = 3 + 25 = 28$ and $D(0) = 4 + 25 = 29$.

Checks

- With $p = q = 1$, part (i) should agree with the main result at $a = \frac{\pi}{2}$, which gives $2I = \frac{\pi}{2} + \ln 1$, so $I = \frac{\pi}{4}$. The formula gives $\frac{\pi/2+0}{2} = \frac{\pi}{4}$. It matches.
- As $a \to \frac{3}{4}\pi$, $\ln(\sin a + \cos a) \to -\infty$. That fits the integrand of I being large and negative just before the denominator vanishes, since $\cos x < 0$ there.
- Numerically, part (ii) is $\frac{2\pi}{25} + \frac{3}{25}\ln\frac{28}{29} \approx 0.247116$, and Simpson's rule on the integral agrees to twelve decimal places. The main result and part (i) also agree with quadrature for 200 random values of a, and of p and q, respectively.

Common slips

- Getting $\ln(q/p)$ instead of $\ln(p/q)$ in part (i). At the upper limit $\frac{\pi}{2}$ it is the sin term that survives, giving p.
- In part (ii), forgetting the constant term and matching only $\sin x$ and $\cos x$. It happens to work, but you should check it rather than assume it.

Beyond STEP

Writing the numerator as "A times the denominator plus B times its derivative" is a small piece of linear algebra. The functions $\sin x$ and $\cos x$ span a two-dimensional

space, and D and D' form another basis of it whenever $p^2 + q^2 \neq 0$. The same idea, finding the right basis to expand in, drives partial fractions and the method of undetermined coefficients for differential equations, which you will meet in a first-year methods course. Siklos's *Advanced Problems in Mathematics* has more integrals that give way to this kind of symmetry trick.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.